Isaac (Zac) Adjei

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[Date]

[Hiring Manager Name/Recruitment Team]
[Company Name]
[Company Address/Location]

Dear [Hiring Manager Name/Recruitment Team] ,

I am writing to apply for the [Role Title] position at [Company Name] . As an Electronic Engineering and Computer Science student at Aston University (Predicted First Class) and Top 40 Finalist in the Black Heritage Undergraduate of the Year Award 2026, I have spent the past three years designing and building embedded hardware systems: from bare-metal microcontroller firmware to multi-node IoT platforms with real-time telemetry processing.

My deepest embedded project is PHAEMOS, an end-to-end IoT predictive maintenance platform I designed and built from scratch. The hardware layer spans four microcontroller families: ESP32 nodes posting telemetry over Wi-Fi, STM32 nodes running HAL C with hardware FFT for vibration analysis, Arduino Nano peripherals and a Raspberry Pi Pico 2W gateway. The firmware handles hardware interrupts, ADC sampling, UART and I2C peripherals and non-blocking state machines that keep latency below the 200ms threshold required by the FastAPI backend. Alongside PHAEMOS I wrote avr-zac, 1,200 lines of bare-metal ATmega644P C with zero external libraries: a 9-mode state machine, hardware interrupt-driven events, PWM output, ADC reads and a watchdog timer, all without an OS or HAL abstraction. Other hardware projects include a 4×4×4 NeoPixel LED cube (64 WS2812B LEDs, LDR-adaptive brightness, non-blocking timing) and a CNC control system with an 8-state FSM, hardware E-Stop interrupt, Schmitt trigger debouncing and a watchdog. I am drawn to [Company Name] because of [specific reason] and I am keen to contribute to [specific area or product] .

Beyond firmware I have analogue hardware depth that complements my digital work. I designed a two-stage TL071/OPA551 audio amplifier from first principles: circuit simulation in Proteus, custom PCB layout, bench fabrication and frequency response verification against a 281mW output target. My Electronic Engineering degree covers filter design, signal processing, power electronics and control systems, giving me the mathematical foundation to debug at the signal level rather than treating hardware as a black box. I also completed a virtual engineering operations placement with British Airways, where I applied data-driven forecasting to A320 maintenance schedules and produced compliance reports aligned with EASA/CAA standards.

I am looking for a role where the firmware genuinely matters: where the gap between a bug and a field failure is real and the code must be correct by design. [Company Name] 's work on [specific product area or technology] is exactly the kind of environment where I want to develop. I write tested, documented embedded C and I am comfortable with oscilloscopes, logic analysers and JTAG workflows as part of a normal working day.

Thank you for considering my application. I would welcome a conversation about how my embedded systems experience can contribute to [Company Name].

Yours sincerely,

Isaac (Zac) Adjei